Admissibility-Preserving Control for Block Strict-Feedback Systems Under Time-Varying Asymmetric Actuator and State Constraints

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***Abstract*— This paper addresses the tracking control of block strict-feedback nonlinear systems subject to time-varying asymmetric actuator and full state constraints. To that end, we develop a time-varying *Admissibility-Preserving Control* (APC) framework, in which the *Admissibility-Preserving Input Realization* (APIR) generates the plant input through a dynamic realization. The APIR guarantees input admissibility under the moving actuator set whenever each actuator bound contracts slower than the decay rate of the realization, and a closed-form lower bound on its gain is derived. The state constraints are handled by a logarithmic barrier coordinate at every backstepping step, with widths measured from the virtual control to the moving envelope of the corresponding state. We prove that every state lies inside its envelope, with each virtual control only required to remain inside the envelope of the next state, while the tracking error converges to zero exponentially. Simulations demonstrate the efficacy of the proposed safety-critical control scheme.**

***Index Terms*— Admissibility-preserving control, backstepping, full state constraints, time-varying constraints, asymmetric actuator limits, safety-critical control.**

I. Introduction

In many practical control systems, actuator limits and state constraints vary over time owing to factors such as actuator derating, faults, and changing operating conditions. Although the design of feedback controllers for constrained nonlinear systems has attracted sustained research attention for over three decades, the systematic treatment of such time-varying constraints remains an open problem. A foundational body of work addresses actuator limits through bounded-input stabilization, in which the feedback is constructed so that the control input remains within prescribed bounds. For linear plants, bounded globally stabilizing laws exist exactly when the plant is asymptotically null controllable [1], while nested-saturation and low-gain designs achieve global stabilization of chains of integrators [2] and semiglobal stabilization of general null-controllable plants [3]. A comprehensive treatment of analysis and synthesis under actuator saturation is given in [4]. Anti-windup designs follow a complementary approach, in which the nominal controller is retained, and an additional compensator modifies the closed-loop dynamics once the actuator saturates [5], [6]. For nonlinear plants, the saturation is usually absorbed by an auxiliary system or by an adaptive compensator placed around the nominal design [7], [8]. However, these designs assume time-invariant actuator bounds, which are mostly symmetric. Moreover, in the anti-windup and auxiliary-system designs, the correction is applied after the nominal design is complete. Consequently, the saturation limit is treated as a constant actuator parameter. In contrast, the fault-tolerant control literature studies authority that changes during operation, where the loss of effectiveness is usually modeled as a multiplicative gain to be estimated and compensated [9].

Barrier Lyapunov functions (BLFs) provide an alternative mechanism for enforcing constraints in strict-feedback systems by adding a term to the Lyapunov function that diverges at the boundary. Consequently, the output/state remains inside its admissible set as long as the Lyapunov function remains bounded. In [10], barrier functions were applied to systems with multiple state constraints, while the authors in [11] developed a BLF-based backstepping design for output-constrained strict-feedback systems, using both symmetric and asymmetric barriers. This framework was subsequently extended to partial state constraints [12], to full state constraints in pure-feedback systems [13] and under unknown control direction [14], to output envelopes that vary in time [15], to time-varying full state constraints through asymmetric barriers [16], and further to networked settings with moving state constraints [17].

However, in all of these designs, the barrier acts on the error between each state and its virtual control. Keeping this error bounded confines the state only if the virtual control itself remains within a reduced region of the state envelope. The resulting requirement is the feasibility condition, which is usually verified offline by searching for design parameters [13], [18]. Integral BLFs place the barrier on the state itself and reduce the conservatism of this condition while retaining it [19], [20]. The authors in [18] removed the feasibility condition through a nonlinear state-dependent transformation combined with dynamic surface control. However, the tracking

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error in [18] is only shown to be uniformly ultimately bounded.

Moreover, prescribed-performance and funnel designs confine the tracking error to a prescribed envelope by transforming the constrained error into an unconstrained coordinate [21], [22]. Since the envelope is a design function of time, time-varying constraints on the tracking error are handled directly. However, these designs constrain only the tracking error, and the input constraints require an additional mechanism. In funnel control, the input constraints were treated as hard constraints in [23], where the funnel boundary is widened by a dynamic law whenever the input saturates and may grow unbounded if the saturation persists. Within prescribed performance control, the authors in [24] handled the input constraints by modifying the reference trajectory, while [25], [26] relaxed the performance specifications adaptively according to the actuator limits, with the boundedness of the closed-loop signals established only on compact sets unless the system is input-to-state stable. The authors in [27] retained both the performance and the input constraints, provided that a feasibility condition relating the prescribed performance to the input bounds holds. In these approaches, the input remains admissible either because the tracking objective is relaxed whenever the actuator authority is insufficient, or because the performance specification is restricted a priori.

Along similar lines, control barrier functions (CBFs) provide a mechanism for enforcing state constraints by rendering a safe set forward invariant. In [28], CBFs were unified with control Lyapunov functions through a quadratic program (QP) solved online, which minimally modifies a nominal control law to retain safety. This framework was subsequently extended to constraints of high relative degree [29]. However, the input bounds enter the QP as additional constraints, and the QP may become infeasible when these bounds conflict with the CBF conditions [29]. The authors in [30] introduced adaptive CBFs that accommodate time-varying control bounds and guarantee the feasibility of the QP, provided that the original optimal control problem is feasible. Nevertheless, in these designs, the input bounds are satisfied only through the constraints of the online QP, so that input admissibility depends on the feasibility of the program at every time instant. Reference and command governors [31] also retain the nominal control law, and enforce the constraints by modifying the reference applied to the closed loop.

The Admissibility-Preserving Control (APC) framework has been recently introduced in our earlier work for a time-invariant asymmetric actuator set and a time-varying output constraint [32]. APC generates the physical input through a differential equation, termed the admissibility-preserving input realization (APIR), whose effective gain vanishes at the actuator bounds. Therein, the admissible input set is forward invariant by construction, and the realized input is continuously differentiable, which allows it to be treated as an additional state in a recursive Lyapunov design. This framework has since been extended to networked coordination [33], and the constrained-motion applications that motivate it are described in [34].

However, three gaps remain when the admissible sets vary in time. First, the forward invariance of a time-invariant actuator set does not carry over to a moving one. Second, BLF backstepping [12], [13], [16] confines each state only if its virtual control keeps the barrier width away from the envelope, a feasibility condition that is verified offline [13], [18]. Third, since the command of the APIR is formed by dividing by the realization gain [32], this gain cancels in the closed loop, and an explicit compatibility condition that bounds the command is required. To the best of our knowledge, tracking control of nonlinear systems under time-varying asymmetric actuator and full-state constraints, with both certified by a single explicit condition, remains unaddressed. We close this gap by extending APC to moving admissible sets for block strict-feedback systems. We summarize our contributions below.

- A time-varying asymmetric APIR is developed whose moving actuator set is forward invariant under bounded commands, provided that each actuator bound contracts slower than the decay rate of the realization. The analysis in a normalized coordinate also establishes a compact invariant interior set and a closed-form lower bound on the APIR gain.
- A logarithmic barrier coordinate is applied at every block of the backstepping recursion, with its widths measured from the virtual control to the moving envelope of the corresponding state. Unlike BLF backstepping [12], [13], [16], the full state constraints hold without barrier widths reserved for the errors, and the feasibility condition is replaced by a separation condition that only requires each virtual control to lie inside the envelope of the next state. This separation condition and the boundedness of the commanded input are combined into a single APC compatibility test on a Lyapunov sublevel set.
- A closed-form ratio between each error and its barrier coordinate is derived, which carries the cross terms between consecutive coordinates. Consequently, the Lyapunov identity of the closed-loop system is exact and yields an explicit decay rate.
- Under APC compatibility, we show that the tracking error converges exponentially to zero with an explicit rate and amplitude, while all state envelopes and the moving actuator set remain forward invariant and all closed-loop signals remain bounded. Moreover, provided that the APIR is designed on a correspondingly tightened actuator set that satisfies the boundary-rate condition, input admissibility transfers to the physical actuator when the applied input deviates from the realized input by a bounded error.

II. Problem Formulation

Throughout this paper, $\mathbb{R}^m$ denotes the m-dimensional Euclidean space, $\mathbb{R}_+$ the set of positive reals, and I, $\mathbf{1}$, and $\mathbf{0}$ the identity matrix and vectors of ones and zeros, respectively, of appropriate dimensions. The transpose and absolute value are denoted by $(\cdot)^\top$ and $|\cdot|$, respectively. Vector inequalities, products, and quotients are understood componentwise. For square matrices, $A \succeq B$ indicates that $A - B$ is positive semidefinite. The operators $\mathrm{diag}(\cdot)$ and $\sigma_{\min}(\cdot)$ denote a diagonal matrix and the smallest singular value, respectively. The Euclidean

and infinity norms are denoted by $\|\cdot\|$ and $\|\cdot\|_\infty$, respectively, and $\mathrm{dist}_\infty(v,\mathbb{K}) := \inf_{w\in\mathbb{K}}\|v-w\|_\infty$ denotes the infinity-norm distance to $\mathbb{K}\subset\mathbb{R}^m$. A time-varying set $\mathbb{S}(t)$ is forward invariant if $x(0)\in\mathbb{S}(0)$ implies $x(t)\in\mathbb{S}(t)\ \forall t\geq 0$.

Consider the class of block strict-feedback nonlinear systems given by

$$\dot{x}_i = f_i(\bar{x}_i) + G_i(\bar{x}_i)x_{i+1}, \quad i = 1,\dots,n-1, \tag{1a}$$
$$\dot{x}_n = f_n(\bar{x}_n) + G_n(\bar{x}_n)u, \tag{1b}$$
$$y = x_1, \tag{1c}$$

where $x_i\in\mathbb{R}^m$ is the ith state block, $\bar{x}_i = [x_1^\top,\dots,x_i^\top]^\top$, $u\in\mathbb{R}^m$ is the input applied to the plant, $y\in\mathbb{R}^m$ is the measured output, and $f_i:\mathbb{R}^{im}\to\mathbb{R}^m$ and $G_i:\mathbb{R}^{im}\to\mathbb{R}^{m\times m}$ are smooth. Note that the dynamics of x_i are affine in x_{i+1}, which acts as the input to the ith subsystem. In other words, (1) is control affine at every step of the recursion.

For a safe operation, each state block x_i and the plant input u are required to remain inside the time-varying sets

$$\mathbb{X}_i(t) := \left\{x\in\mathbb{R}^m \mid \underline{x}_i(t) < x < \bar{x}_i(t)\right\}, i = 1,\dots,n, \tag{2}$$
$$\mathbb{U}(t) := \left\{u\in\mathbb{R}^m \mid u_{\min}(t) < u < u_{\max}(t)\right\}, \tag{3}$$

with $u_{\min}(t) < 0 < u_{\max}(t)$ for all $t\geq 0$. Since $y = x_1$, $\mathbb{X}_1(t)$ also serves as the output constraint. The sets are illustrated for $m = 2$ in Figure 1.

Assumption 1 (Model regularity and control effectiveness). For each i, the maps f_i and G_i are known, smooth, and available for synthesis, and $G_i(\bar{x}_i)$ is invertible with $\sigma_{\min}(G_i(\bar{x}_i)) \geq \underline{g}_i > 0$ for all $\bar{x}_i$, where $\underline{g}_i$ is a known constant.

The bound $\underline{g}_i$ keeps every block uniformly invertible, so no direction of x_i loses actuation through x_{i+1}. No upper bound is imposed, admitting input gains that grow with the state.

Assumption 2. The reference y_d is $n+1$ times continuously differentiable with bounded derivatives, and there exists $\varepsilon_y > 0$ such that $\underline{x}_1(t) + \varepsilon_y\mathbf{1} \leq y_d(t) \leq \bar{x}_1(t) - \varepsilon_y\mathbf{1}$ for all $t\geq 0$.

Owing to ε_y, the reference lies strictly inside the envelope, and hence exact tracking itself is an admissible motion.

Assumption 3 (State envelopes). The bounds $\underline{x}_1$ and $\bar{x}_1$ are $n+1$ times continuously differentiable, and for $i = 2,\dots,n$ the bounds $\underline{x}_i$ and $\bar{x}_i$ are $n+2-i$ times continuously differentiable. All of these derivatives are bounded, the bounds themselves are bounded, and $\underline{x}_i(t) < \bar{x}_i(t)$ for all $t\geq 0$ and every i.

Envelopes generated by a scheduling law or by a reference model are smooth functions of time with bounded derivatives and satisfy Assumption 3. The strict inequality $\underline{x}_i(t) < \bar{x}_i(t)$ keeps every set $\mathbb{X}_i(t)$ nonempty for all $t\geq 0$. We are now in a position to formally define our problem statement.

Problem 1 (Admissibility-preserving constrained tracking)**.** *Consider the system* (1) *with the time-varying sets* (2) *and* (3)*. Develop a control law such that, for every initial condition with* $x_i(0)\in\mathbb{X}_i(0)$, $i = 1,\dots,n$, *and* $u(0)\in\mathbb{U}(0)$ *that is compatible with the available actuator authority, the following hold for all* $t\geq 0$*:*

1) *every state block satisfies* $x_i(t)\in\mathbb{X}_i(t)$, $i = 1,\dots,n$*;*
2) *the realized plant input satisfies* $u(t)\in\mathbb{U}(t)$*;*
3) *the tracking error* $y(t) - y_d(t)$ *converges to zero exponentially, and all closed-loop signals remain bounded.*

Note that Problem 1 is posed on compatible initial conditions and not globally, since, with finite authority, an initial state far from the reference, or a demanding reference, cannot be steered by any controller without violating (2) or (3). Moreover, the first requirement is stronger than an output constraint, since every intermediate state block must remain inside its moving envelope. Such a requirement arises, for instance, in flight vehicles, where the angle of attack and body rates must remain within envelopes that vary with flight conditions [35], [36], and in robotic manipulators [37], where joint velocities are limited in addition to joint positions.

III. The Time-Varying APC Framework

This section develops time-varying APC through three elements: *APIR realizes admissibility under a moving actuator set*, *APC synthesis achieves the control objective around the admissible realization*, and *APC compatibility certifies feasible use of finite authority*. We first establish the admissibility and gain properties of the APIR under moving bounds. The realization is then incorporated into backstepping, with a barrier coordinate transformation at every block so that each state remains inside its moving envelope. Finally, the APC compatibility condition is stated, under which every constraint is met with the available actuator authority.

A. Admissibility-Preserving Input Realization under Moving Bounds

The APIR is first analyzed as an independent dynamic module, whose admissibility depends only on the boundedness of the commanded input and on the rate at which the actuator bounds move, and not on the plant. In this paper, the APIR of [32] is extended to the moving actuator set (3) through the componentwise realization

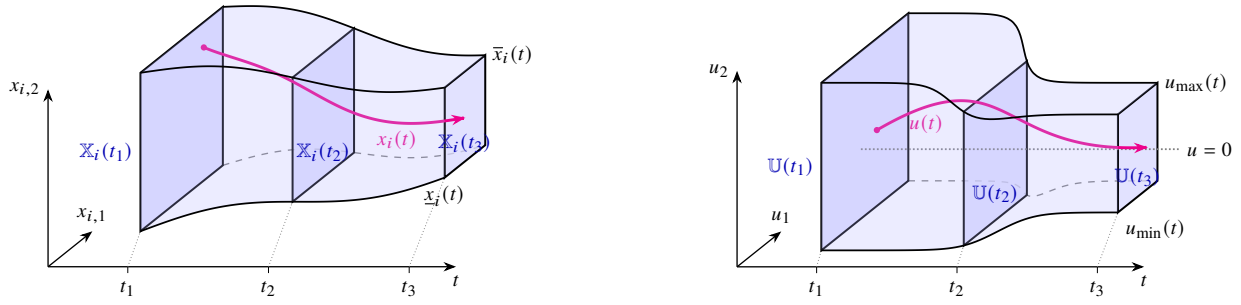
$$\dot{u} = \mathcal{G}(u,t)u_c - \mathcal{F}(u), \tag{4}$$

where u_c is the commanded input generated by the synthesis layer, u is the realized actuator output supplied to the plant, and $\mathcal{G}(u,t)$ and $\mathcal{F}(u)$ are given by

$$\mathcal{G}(u,t) := p_1\,\mathrm{diag}\left(\mathcal{S}_l(u_l,t)\right), \mathcal{F}(u) := p_1p_2u, \tag{5}$$

$$\mathcal{S}_l(u_l,t) := \begin{cases} 1-\left(\dfrac{u_l}{u_{\max,l}(t)}\right)^\gamma, & u_l > 0, \\ 1-\left(\dfrac{u_l}{u_{\min,l}(t)}\right)^\gamma, & u_l \leq 0, \end{cases} \tag{6}$$

where $p_1,p_2\in\mathbb{R}_+$ are design parameters, γ is an even positive integer that determines the smoothness order and transition profile of the realization, and $u_{\min,l}(t) < 0 < u_{\max,l}(t)$ are the lower and upper bounds of the lth actuator channel. Since both branches of (6) yield $\mathcal{S}_l = 1$ at $u_l = 0$, the APIR in (4) is locally Lipschitz in u. In contrast to [32], the bounds are time-varying, and hence $\mathcal{G}$ depends explicitly on time. A parametric sensitivity study of the APIR is reported in [32].



(a) State set. (b) Input set.

Fig. 1: Time-varying state and input sets.

Assumption 4 (Admissible actuator bounds and boundary rates). The bounds $u_{\max}(t)$ and $u_{\min}(t)$ are continuously differentiable, and there exist constants $\underline{u}_{\max} > 0$, $\underline{u}_{\min} > 0$, and $\varepsilon_0 \in (0, p_1 p_2)$ such that, for every component l and all $t \geq 0$,

$$u_{\max,l}(t) \geq \underline{u}_{\max}, u_{\min,l}(t) \leq -\underline{u}_{\min}, \tag{7}$$

$$\frac{\dot{u}_{\max,l}(t)}{u_{\max,l}(t)} \geq -p_1 p_2 + \varepsilon_0, \frac{\dot{u}_{\min,l}(t)}{u_{\min,l}(t)} \geq -p_1 p_2 + \varepsilon_0. \tag{8}$$

Note that (8) constrains the logarithmic derivatives of $u_{\max,l}$ and $|u_{\min,l}|$, that is, their instantaneous exponential rates of change. Therefore, (8) allows each actuator boundary to contract only at an exponential rate below $p_1 p_2$, which is the rate at which the term $\mathcal{F}(u) = p_1 p_2 u$ in (5) pulls the realized input toward the origin, whereas expansion is in general unrestricted. The margin ε_0 must be uniform in time, since a pointwise strict inequality admits ratios that approach $-p_1 p_2$ asymptotically, under which the confinement margin degenerates. The condition in (7) ensures that the available authority does not vanish asymptotically, which keeps the ratio between the command bound and the available authority uniformly bounded.

Theorem 1 (Intrinsic admissibility and compact interiority under moving bounds)**.** *Consider* (4) *under Assumption 4, with a piecewise-continuous command satisfying* $\|u_c(t)\|_\infty \leq \xi$ *for all* $t \geq 0$. *Define*

$$K_{\max} := \frac{p_1 \xi}{\underline{u}_{\max}}, K_{\min} := \frac{p_1 \xi}{\underline{u}_{\min}}, \tag{9}$$

and let $s^*, \zeta^* \in (0,1)$ *satisfy*

$$K_{\max}\,[1 - (s^*)^\gamma] < s^* \varepsilon_0, K_{\min}\,[1 - (\zeta^*)^\gamma] < \zeta^* \varepsilon_0. \tag{10}$$

For every $u(0) \in \mathbb{U}(0)$, *define*

$$\bar{s} := \max\left\{s^*, \max_l \frac{u_l(0)}{u_{\max,l}(0)}\right\},$$
$$\bar{\zeta} := \max\left\{\zeta^*, \max_l \frac{u_l(0)}{u_{\min,l}(0)}\right\}. \tag{11}$$

Then $\bar{s}, \bar{\zeta} \in (0,1)$, *and the solution of* (4) *exists for all* $t \geq 0$ *and satisfies*

$$\bar{\zeta} u_{\min}(t) \leq u(t) \leq \bar{s} u_{\max}(t), \qquad t \geq 0, \tag{12}$$

Consequently, $\mathbb{U}(t)$ *is forward invariant, and the realization gain satisfies*

$$\mathcal{G}(u(t), t) \geq \underline{\mathcal{G}} I, \underline{\mathcal{G}} := p_1 \left[1 - \max\{\bar{s}, \bar{\zeta}\}^\gamma\right] > 0. \tag{13}$$

Proof. Note that s^* and ζ^* exist, since, as $s^* \to 1$ and $\zeta^* \to 1$, the left sides of (10) tend to zero, whereas the right sides tend to $\varepsilon_0 > 0$. As (4) is diagonal, it suffices to consider a single component, whose index is suppressed. Local existence and uniqueness follow from the local Lipschitz continuity of (4) in u and its piecewise continuity in t. Define the normalized coordinate $s := u/u_{\max}$, which is well defined by (7). For $u > 0$, one has $\mathcal{S} = 1 - s^\gamma$ and $u = s u_{\max}$, so differentiating s along (4) yields

$$\dot{s} = \frac{\dot{u}}{u_{\max}} - s\frac{\dot{u}_{\max}}{u_{\max}} = \frac{p_1\,[1 - s^\gamma]\,u_c - p_1 p_2 s u_{\max}}{u_{\max}} - s\frac{\dot{u}_{\max}}{u_{\max}}$$
$$= \frac{p_1 u_c}{u_{\max}}\,[1 - s^\gamma] - s\lambda, \tag{14}$$

where $\lambda := p_1 p_2 + \dot{u}_{\max}/u_{\max}$ satisfies $\lambda(t) \geq \varepsilon_0$ for all $t \geq 0$ by (8). Since $1 - s^\gamma > 0$ for $s \in (0,1)$, $u_c \leq \xi$, and $u_{\max} \geq \underline{u}_{\max}$ by (7), the first term in (14) is bounded above by $K_{\max}[1 - s^\gamma]$. Together with $\lambda \geq \varepsilon_0$, this yields

$$\dot{s} \leq K_{\max}\,[1 - s^\gamma] - s\varepsilon_0, \quad s \in (0,1). \tag{15}$$

Now, let $s \in [s^*, 1)$. As $s \mapsto 1 - s^\gamma$ is decreasing on $[0,1)$, one has $K_{\max}[1 - s^\gamma] \leq K_{\max}[1 - (s^*)^\gamma]$. Moreover, (10) gives $K_{\max}[1 - (s^*)^\gamma] < s^*\varepsilon_0$, and $s^*\varepsilon_0 \leq s\varepsilon_0$ since $s \geq s^*$. Combining these inequalities yields $K_{\max}[1 - s^\gamma] < s\varepsilon_0$, so the right side of (15) is negative. Therefore, $\dot{s} < 0$ for all $s \in [s^*, 1)$.

Since $\bar{s} \in [s^*, 1)$, one has $\dot{s} < 0$ whenever $s = \bar{s}$, so s cannot cross $\bar{s}$ from below. As $s(0) \leq \bar{s}$ by (11), it follows that $s(t) \leq \bar{s}$ for all $t \geq 0$. Multiplying by $u_{\max}(t) > 0$ yields the upper inequality in (12).

The lower inequality follows similarly. Define $\zeta := u/u_{\min}$, which is well defined by (7) and positive exactly when $u < 0$. For $u < 0$, one has $\mathcal{S} = 1 - \zeta^\gamma$ and $u = \zeta u_{\min}$, so differentiating ζ along (4) yields $\dot{\zeta} = (p_1 u_c/u_{\min})[1 - \zeta^\gamma] - \zeta(p_1 p_2 + \dot{u}_{\min}/u_{\min})$, where $p_1 p_2 + \dot{u}_{\min}/u_{\min} \geq \varepsilon_0$ for all $t \geq 0$ by (8). Since $1 - \zeta^\gamma > 0$ for $\zeta \in (0,1)$, $u_c/u_{\min} \leq \xi/|u_{\min}|$ because $u_c \geq -\xi$ and $u_{\min} < 0$, and $|u_{\min}| \geq \underline{u}_{\min}$ by (7), the first term is bounded above by $K_{\min}[1 - \zeta^\gamma]$. Together with (8), this yields $\dot{\zeta} \leq K_{\min}[1 - \zeta^\gamma] - \zeta\varepsilon_0$ for $\zeta \in (0,1)$. For $\zeta \in [\zeta^*, 1)$, since $\zeta \mapsto 1 - \zeta^\gamma$ is decreasing and $\zeta \geq \zeta^*$, (10) gives $K_{\min}[1 - \zeta^\gamma] \leq K_{\min}[1 - (\zeta^*)^\gamma] < \zeta^*\varepsilon_0 \leq \zeta\varepsilon_0$, so $\dot{\zeta} < 0$. Since $\bar{\zeta} \in [\zeta^*, 1)$, ζ cannot cross $\bar{\zeta}$ from below. As $\zeta(0) \leq \bar{\zeta}$ by (11), it follows that $\zeta(t) \leq \bar{\zeta}$ for all $t \geq 0$. Multiplying by $u_{\min}(t) < 0$ yields lower inequality in (12).

Finally, by (5) and (6), each diagonal entry of $\mathcal{G}$ equals $p_1[1-s^\gamma]$ when $u>0$ and $p_1[1-\zeta^\gamma]$ when $u\le 0$. Since $0<s\le\bar{s}$ in the first case and $0\le\zeta\le\bar{\zeta}$ in the second, the entry is at least $p_1[1-\max\{\bar{s},\bar{\zeta}\}^\gamma]$, which gives (13). Since u remains bounded, local Lipschitz continuity yields forward completeness. □

Remark 1. Note that the normalized coordinates s and ζ also yield the gain bound (13) in closed form. In [32], the corresponding bound for fixed actuator limits was obtained by minimizing a continuous function over a compact invariant interval, so the resulting constant was not available explicitly. In contrast, $\underline{\mathcal{G}}$ in (13) follows directly from $\bar{s}$ and $\bar{\zeta}$, which are computed from (10) and (11). In addition, (8) can be understood as a rate budget for authority loss, so the product p_1p_2 sets both the equilibrium margin of the realization and the fastest admissible contraction of the actuator bounds.

Remark 2. Let $\gamma=2$, and recall $K_{\max}$ and $K_{\min}$ from (9) and the rate margin ε_0 from (8). Then the first condition in (10) is equivalent to $K_{\max}(s^*)^2+\varepsilon_0 s^*-K_{\max}>0$. The left side equals $-K_{\max}<0$ at $s^*=0$ and $\varepsilon_0>0$ at $s^*=1$, so this quadratic has exactly one root in $(0,1)$, given by

$$s_+ \coloneqq \frac{\sqrt{\varepsilon_0^2+4K_{\max}^2}-\varepsilon_0}{2K_{\max}}, \tag{16}$$

and the condition holds if and only if $s^*\in(s_+,1)$. Similarly, the second condition in (10) holds if and only if $\zeta^*\in(\zeta_+,1)$, where ζ_+ is given by (16) with $K_{\min}$ in place of $K_{\max}$. Hence, for $\gamma=2$, the constants $\bar{s}$, $\bar{\zeta}$, and $\underline{\mathcal{G}}$ in (11) and (13) are available in closed form.

B. APIR-Augmented APC for State-Constrained Tracking

We now interconnect the APIR (4) with the plant (1) to solve Problem 1. To enforce the state constraints, we apply a smooth asymmetric logarithmic barrier coordinate at every block, measured from a reference signal to the lower and upper boundaries of the corresponding envelope. At the first block, the reference is y_d, whereas at block $i\ge 2$, it is the virtual control that the preceding step assigns to x_i. Now, the barrier widths become the distances from that virtual control to the envelope of x_i, and a finite coordinate is equivalent to admissibility of the state itself.

To that end, we define the errors

$$\varphi_1 \coloneqq x_1-y_d,\ \varphi_i \coloneqq x_i-\eta_{i-1},\ \varrho \coloneqq u-\eta_n, \tag{17}$$

for $i=2,\dots,n$, where η_i, $i=1,\dots,n-1$, is the virtual control input for x_{i+1}, and η_n is the virtual input demanded of the APIR. The corresponding barrier widths are defined as

$$\alpha_1 \coloneqq y_d-\underline{x}_1,\ \beta_1 \coloneqq \overline{x}_1-y_d, \tag{18}$$

$$\alpha_i \coloneqq \eta_{i-1}-\underline{x}_i,\ \beta_i \coloneqq \overline{x}_i-\eta_{i-1},\ i=2,\dots,n. \tag{19}$$

Note that $\alpha_1\succ 0$ and $\beta_1\succ 0$ by Assumption 2. For $i=1,\dots,n$, a componentwise barrier coordinate transformation is defined as

$$z_{i,l} \coloneqq \ln\!\left(\frac{\beta_{i,l}[\alpha_{i,l}+\varphi_{i,l}]}{\alpha_{i,l}[\beta_{i,l}-\varphi_{i,l}]}\right). \tag{20}$$

Since the widths $\alpha_{i,l}$ and $\beta_{i,l}$ vary with time, $z_{i,l}$ changes both through the error $\varphi_{i,l}$ and through the widths. Writing (20) as $z_{i,l}=\ln\beta_{i,l}-\ln\alpha_{i,l}+\ln(\alpha_{i,l}+\varphi_{i,l})-\ln(\beta_{i,l}-\varphi_{i,l})$ and differentiating it, with the indices suppressed, gives $\dot{z}=\dot{\beta}/\beta-\dot{\alpha}/\alpha+(\dot{\alpha}+\dot{\varphi})/(\alpha+\varphi)-(\dot{\beta}-\dot{\varphi})/(\beta-\varphi)$, which, on collecting terms, becomes $\dot{z}_{i,l}=q_{i,l}\dot{\varphi}_{i,l}+\psi_{i,l}$, where where

$$q_{i,l} \coloneqq \frac{\partial z_{i,l}}{\partial\varphi_{i,l}}=\frac{1}{\alpha_{i,l}+\varphi_{i,l}}+\frac{1}{\beta_{i,l}-\varphi_{i,l}}>0, \tag{21}$$

$$\psi_{i,l} \coloneqq \dot{\alpha}_{i,l}\left(\frac{1}{\alpha_{i,l}+\varphi_{i,l}}-\frac{1}{\alpha_{i,l}}\right)+\dot{\beta}_{i,l}\left(\frac{1}{\beta_{i,l}}-\frac{1}{\beta_{i,l}-\varphi_{i,l}}\right). \tag{22}$$

Here, $q_{i,l}$ is the rate at which $z_{i,l}$ changes with the error $\varphi_{i,l}$ when the widths are held fixed, whereas $\psi_{i,l}$ accounts for the effect of the moving widths and vanishes when $\dot{\alpha}_{i,l}=\dot{\beta}_{i,l}=0$. Stacking the components with $Q_i \coloneqq \operatorname{diag}(q_{i,l})$ and $\psi_i \coloneqq [\psi_{i,1},\dots,\psi_{i,m}]^\top$ gives

$$\dot{z}_i=Q_i\dot{\varphi}_i+\psi_i. \tag{23}$$

For $i\ge 2$, the widths move with the virtual control, since $\dot{\alpha}_i=\dot{\eta}_{i-1}-\dot{\underline{x}}_i$ and $\dot{\beta}_i=\dot{\overline{x}}_i-\dot{\eta}_{i-1}$. Moreover, $\alpha_i+\beta_i=2r_i$, where $\breve{x}_i \coloneqq \frac{1}{2}[\overline{x}_i+\underline{x}_i]$ and $r_i \coloneqq \frac{1}{2}[\overline{x}_i-\underline{x}_i]$ denote the centre and the half-width of $\mathbb{X}_i(t)$.

Lemma 1. *Consider the barrier coordinate transformation (20) with $\alpha_i(t)\succ 0$ and $\beta_i(t)\succ 0$. Then the map $\varphi_i\mapsto z_i$ is a smooth diffeomorphism from $(-\alpha_i,\beta_i)$ onto $\mathbb{R}^m$, and $z_i=0$ if and only if $\varphi_i=0$. Moreover, z_i is finite if and only if $\underline{x}_i(t)<x_i(t)<\overline{x}_i(t)$. Furthermore, the ratio $\kappa_{i,l} \coloneqq \varphi_{i,l}/z_{i,l}$ is given by*

$$\kappa_{i,l}=\frac{\alpha_{i,l}\beta_{i,l}(e^{z_{i,l}}-1)}{z_{i,l}(\beta_{i,l}+\alpha_{i,l}e^{z_{i,l}})}, \tag{24}$$

which extends smoothly to $z_{i,l}=0$ with the value $\alpha_{i,l}\beta_{i,l}/(\alpha_{i,l}+\beta_{i,l})$, and $\kappa_{i,l}>0$ for all $z_{i,l}\in\mathbb{R}$.

Proof. Since (20) acts componentwise, it suffices to fix t and l and to write α, β, φ and z for $\alpha_{i,l}(t)$, $\beta_{i,l}(t)$, $\varphi_{i,l}$ and $z_{i,l}$. On the interval $(-\alpha,\beta)$ both $\alpha+\varphi$ and $\beta-\varphi$ are positive, so (20) can be written as $z=\ln\beta-\ln\alpha+\ln(\alpha+\varphi)-\ln(\beta-\varphi)$, which is smooth in φ, and its derivative is $q_{i,l}>0$ from (21). The map $\varphi\mapsto z$ is strictly increasing and hence injective. As $\varphi\to-\alpha^+$, the term $\ln(\alpha+\varphi)$ tends to $-\infty$ while the remaining terms stay bounded, so $z\to-\infty$. Similarly, as $\varphi\to\beta^-$, the term $-\ln(\beta-\varphi)$ tends to $+\infty$, so $z\to+\infty$. By continuity, the image of $(-\alpha,\beta)$ is all of $\mathbb{R}$. Solving (20) for φ gives the inverse

$$\varphi=\frac{\alpha\beta(e^z-1)}{\beta+\alpha e^z}, \tag{25}$$

whose denominator is positive for every $z\in\mathbb{R}$, so the inverse is smooth as well, and the componentwise map is a smooth diffeomorphism onto $\mathbb{R}^m$. At $\varphi=0$ the argument of the logarithm in (20) equals $\beta\alpha/(\alpha\beta)=1$, so $z=0$, and injectivity excludes any other zero. Outside $(-\alpha,\beta)$ the argument of the logarithm is zero, negative or undefined, so z is finite exactly when $\varphi\in(-\alpha,\beta)$. From (17) and (19), $\alpha_i+\varphi_i=x_i-\underline{x}_i$ and $\beta_i-\varphi_i=\overline{x}_i-x_i$ for $i\ge 2$, and the same holds for $i=1$ by (18), so the admissible interval of φ_i is exactly the envelope of x_i. Dividing (25) by z gives (24). The function $(e^z-1)/z$

extends smoothly to $z = 0$ with the value 1, and it is positive because $e^z - 1$ and z share the same sign. The denominator $\beta+\alpha e^z$ is positive, so $\kappa_{i,l}$ is smooth and positive, and its value at $z = 0$ is $\alpha\beta/(\alpha + \beta)$. □

Note that, by Lemma 1, the admissible interval of φ_i coincides with the envelope of x_i. Hence, unlike BLF backstepping [12], [13], [16], no band of fixed width around the virtual control is reserved, and no allocation of the envelope is needed. However, the coordinate z_{i+1} is defined only while $\alpha_{i+1} \succ 0$ and $\beta_{i+1} \succ 0$, that is, while

$$\underline{x}_{i+1}(t) < \eta_i(t) < \overline{x}_{i+1}(t), \qquad i = 1, \dots, n-1, \tag{26}$$

which is the counterpart of Assumption 2 at block $i + 1$, with the virtual control η_i in place of the reference. Equivalently, $\|(\eta_i - \check{x}_{i+1})/r_{i+1}\|_\infty < 1$. In other words, (26) requires the virtual control only to lie inside the envelope, whereas the feasibility condition of BLF backstepping requires it to keep the barrier width away from the envelope. Moreover, the ratio κ_i in (24) carries the cross term from block $i - 1$ into the coordinate z_i.

We now use *backstepping* based control to design the time-varying APC strategy. The design proceeds in $n + 1$ steps, where the virtual controls $\eta_1, \dots, \eta_n$ are designed in Steps $1, \dots, n$, and the command u_c of the APIR is designed in Step $n + 1$. In what follows, we select the weights $\mu_i > 0$, $i = 1, \dots, n$, and define $R_i := \mathrm{diag}(\kappa_{i,l})$ and the cross terms $\chi_i := \mu_i G_i^\top Q_i z_i$.

Step 1: One can obtain the time derivative of φ_1 using (1) and $x_2 = \varphi_2 + \eta_1$ as

$$\dot{\varphi}_1 = \dot{x}_1 - \dot{y}_d = f_1 + G_1(\varphi_2 + \eta_1) - \dot{y}_d. \tag{27}$$

By viewing x_2 as a virtual control input, let us consider a Lyapunov function candidate $\mathcal{W}_1 = \frac{1}{2}\mu_1\|z_1\|^2$, whose time derivative along the state trajectories results in $\dot{\mathcal{W}}_1 = \mu_1 z_1^\top \dot{z}_1$. If we design the virtual control η_1 as

$$\eta_1 = G_1^{-1}\left[\dot{y}_d - f_1 - Q_1^{-1}(\psi_1 + k_1 z_1)\right], \quad k_1 > 0, \tag{28}$$

then substituting (27) and (28) into (23) with $i = 1$ yields the closed-loop dynamics, $\dot{z}_1 = Q_1[f_1 + G_1\varphi_2 + G_1\eta_1 - \dot{y}_d] + \psi_1 = Q_1[G_1\varphi_2 - Q_1^{-1}(\psi_1 + k_1 z_1)] + \psi_1 = -k_1 z_1 + Q_1 G_1 \varphi_2$, such that the derivative of $\mathcal{W}_1$ becomes

$$\begin{aligned}\dot{\mathcal{W}}_1 &= -\mu_1 k_1 \|z_1\|^2 + \mu_1 z_1^\top Q_1 G_1 \varphi_2 \\ &= -\mu_1 k_1 \|z_1\|^2 + \chi_1^\top \varphi_2,\end{aligned} \tag{29}$$

where the last equality follows from $Q_1 = Q_1^\top$ and the definition of χ_1. Note that the cross term $\chi_1^\top\varphi_2$ in (29) is sign indefinite, and it is cancelled in the next step.

Step i ($2 \le i \le n$): In the ith step, the error $\varphi_i = x_i - \eta_{i-1}$, on differentiating with respect to time and using (1), yields

$$\dot{\varphi}_i = \dot{x}_i - \dot{\eta}_{i-1} = f_i + G_i(\varphi_{i+1} + \eta_i) - \dot{\eta}_{i-1}, \tag{30}$$

where $\varphi_{n+1} := \varrho$, since the next block at $i = n$ is the realized input. Note that η_{i-1} is a smooth function of $\bar{x}_{i-1}$ and t, so its derivative along (1) is available analytically as

$$\dot{\eta}_{i-1} = \sum_{j=1}^{i-1} \frac{\partial \eta_{i-1}}{\partial x_j}\left(f_j + G_j x_{j+1}\right) + \frac{\partial \eta_{i-1}}{\partial t}, \tag{31}$$

which enters both (30) and, through $\dot{\alpha}_i$ and $\dot{\beta}_i$, the function ψ_i. By considering x_{i+1} a virtual control input to the ith subsystem, we choose a Lyapunov function candidate as $\mathcal{W}_i = \mathcal{W}_{i-1} + \frac{1}{2}\mu_i\|z_i\|^2$. Its time derivative contains the cross term $\chi_{i-1}^\top\varphi_i$ left by the preceding step, as in (29) for $i = 2$. Since $\varphi_i = R_i z_i$ by Lemma 1, this term can be written as $\chi_{i-1}^\top\varphi_i = \chi_{i-1}^\top R_i z_i$. The virtual control η_i is therefore designed as

$$\eta_i = G_i^{-1}\left[\dot{\eta}_{i-1} - f_i - Q_i^{-1}\left(\psi_i + k_i z_i + \mu_i^{-1} R_i \chi_{i-1}\right)\right], \tag{32}$$

where $k_i > 0$, which, on substituting (30) and (32) into (23), yields the closed-loop dynamics

$$\begin{aligned}\dot{z}_i &= Q_i\left[f_i + G_i\varphi_{i+1} + G_i\eta_i - \dot{\eta}_{i-1}\right] + \psi_i \\ &= Q_i G_i \varphi_{i+1} - \left(\psi_i + k_i z_i + \mu_i^{-1} R_i \chi_{i-1}\right) + \psi_i \\ &= -k_i z_i - \mu_i^{-1} R_i \chi_{i-1} + Q_i G_i \varphi_{i+1}.\end{aligned} \tag{33}$$

By differentiating $\mathcal{W}_i$ with respect to time along the state trajectories and using (33), one has

$$\begin{aligned}\dot{\mathcal{W}}_i &= \dot{\mathcal{W}}_{i-1} + \mu_i z_i^\top \dot{z}_i = -\sum_{j=1}^{i-1}\mu_j k_j \|z_j\|^2 + \chi_{i-1}^\top\varphi_i \\ &\quad - \mu_i k_i \|z_i\|^2 - z_i^\top R_i \chi_{i-1} + \chi_i^\top \varphi_{i+1} \\ &= -\sum_{j=1}^{i}\mu_j k_j \|z_j\|^2 + \chi_i^\top\varphi_{i+1},\end{aligned} \tag{34}$$

where the last equality follows from the relation $z_i^\top R_i \chi_{i-1} = (R_i z_i)^\top \chi_{i-1} = \varphi_i^\top \chi_{i-1}$, since R_i is diagonal. In other words, the virtual control (32) cancels the cross term of the preceding step and adds the damping $-\mu_i k_i \|z_i\|^2$. At $i = n$, (31) expresses $\dot{\eta}_n$ as a function of $\bar{x}_n$ and u.

Step n + 1: In the last step, the realization error is defined as $\varrho = u - \eta_n$, which on differentiating with respect to time and using (4), yields

$$\dot{\varrho} = \mathcal{G}(u, t)u_c - \mathcal{F}(u) - \dot{\eta}_n. \tag{35}$$

Unlike φ_i, the error ϱ requires no barrier coordinate transformation, since the admissibility of u is guaranteed by Theorem 1. Let us consider the Lyapunov function candidate $\mathcal{W} = \mathcal{W}_n + \frac{1}{2}\|\varrho\|^2$. If we design the command u_c as

$$\begin{aligned}u_c &= \mathcal{G}(u, t)^{-1}\mathcal{N}, \\ \mathcal{N} &:= \mathcal{F}(u) + \dot{\eta}_n - \chi_n - k_{n+1}\varrho,\end{aligned} \tag{36}$$

where $k_{n+1} > 0$, then substituting (36) into (35) gives $\dot{\varrho} = \mathcal{N} - \mathcal{F}(u) - \dot{\eta}_n = -\chi_n - k_{n+1}\varrho$. Consequently, using (34) with $i = n$ and $\varphi_{n+1} = \varrho$, the derivative of $\mathcal{W}$ along the state trajectories becomes

$$\begin{aligned}\dot{\mathcal{W}} &= \dot{\mathcal{W}}_n + \varrho^\top\dot{\varrho} \\ &= -\sum_{j=1}^{n}\mu_j k_j\|z_j\|^2 + \chi_n^\top\varrho - \varrho^\top\chi_n - k_{n+1}\|\varrho\|^2 \\ &= -\sum_{j=1}^{n}\mu_j k_j\|z_j\|^2 - k_{n+1}\|\varrho\|^2 \le -2\vartheta\mathcal{W},\end{aligned} \tag{37}$$

where $\vartheta := \min_{1\le j\le n+1} k_j$, and the inequality follows from $\mu_j k_j\|z_j\|^2 \ge \vartheta\mu_j\|z_j\|^2$ and $k_{n+1}\|\varrho\|^2 \ge \vartheta\|\varrho\|^2$. The identity holds wherever the closed-loop vector field is defined.

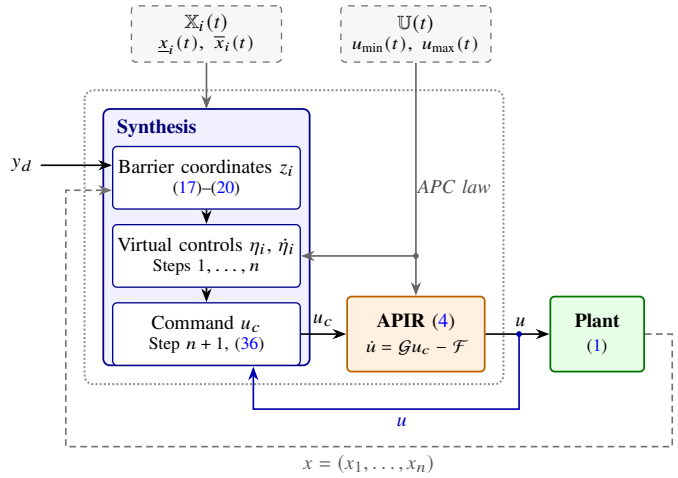


Fig. 2: Time-varying APC architecture.

Note that the weights μ_i do not affect the decay rate in (37), but they scale the cross terms χ_i. When the state and its virtual control lie near the middle of the envelope, one has $\alpha_{i,l} \approx \beta_{i,l} \approx r_{i,l}$, and (20) gives $z_{i,l} \approx 2\varphi_{i,l}/r_{i,l}$. Thus, a wide envelope maps a given physical error into a small z_i, so that equal weights overstate the cross terms between blocks of unequal width. This motivates the choice $\mu_i = \bar{r}_i^2/4$, where $\bar{r}_i$ is a representative half-width of $\mathbb{X}_i$. With this choice, $\frac{1}{2}\mu_i\|z_i\|^2 \approx \frac{1}{2}\|\varphi_i\|^2$, and hence $\mathcal{W}$ approximates half the sum of the squared physical errors of all blocks and the realization.

The term $\dot{\eta}_n$ in (36) depends on every state block, the realized input, the reference, and the envelopes, and its size grows with n. Since η_1 contains $\dot{y}_d$, $\dot{\alpha}_1$, and $\dot{\beta}_1$ and is differentiated n times, the recursion uses $n+1$ derivatives of y_d and of the output envelope. Similarly, the envelope of x_i, $i \geq 2$, enters η_i with its first derivative through ψ_i, and η_i is differentiated $n+1-i$ times, so $n+2-i$ derivatives of that envelope are used, as required by Assumptions 2 and 3.

The resulting architecture of the time-varying APC law is shown in Figure 2. The synthesis block computes the barrier coordinate of each block from its state, virtual control, and envelope, generates the virtual controls in Steps 1 to n, and forms the command u_c in Step $n+1$. The APIR then converts u_c into the admissible input u applied to the plant. Note that the prescribed sets enter the controller as known signals, so the state envelopes and the actuator set are accounted for in the synthesis rather than enforced on its output.

C. APC Compatibility and Closed-Loop Guarantees

To close the dependence between the admissibility of the realization and the boundedness of the command, we introduce the sublevel set $\Omega_c := \{\mathbf{X} \mid \mathcal{W}(\mathbf{X}) \leq c\}$, $c > 0$, where $\mathbf{X} = [z_1^\top, \dots, z_n^\top, \varrho^\top]^\top$. Note that every point of Ω_c, together with t, determines the plant state and the realized input recursively through $x_1 = y_d + R_1 z_1$, $x_{i+1} = \eta_i(\bar{x}_i, t) + R_{i+1} z_{i+1}$, $i = 1, \dots, n-1$, and $u_{\mathbf{X}} = \eta_n(\bar{x}_n, t) + \varrho$. This recursion is well defined as long as (26) holds, since R_{i+1} depends on the widths (19). Moreover, by Lemma 1, every state obtained in this way lies in its envelope. For a fixed margin $\sigma_\eta \in (0,1)$, we call the level c *APC-compatible* if

$$P_c := \sup_{\mathbf{X}\in\Omega_c, t\geq 0} \max_{1\leq i\leq n-1} \left\| \frac{\eta_i(\mathbf{X}, t) - \check{x}_{i+1}(t)}{r_{i+1}(t)} \right\|_\infty \leq 1 - \sigma_\eta, \quad (38)$$

and there exist a compact set $\mathbb{K}_c$, with $\mathbb{K}_c \subset \mathbb{U}(t)$ for all $t \geq 0$, and a constant $\xi > 0$ such that $u_{\mathbf{X}} \in \mathbb{K}_c$ for every $\mathbf{X} \in \Omega_c$ and $t \geq 0$, and

$$Q_c(\xi) := \sup_{\mathbf{X}\in\Omega_c, t\geq 0} \|\mathcal{G}(u_{\mathbf{X}}, t)^{-1} \mathcal{N}(\mathbf{X}, u_{\mathbf{X}}, t)\|_\infty \leq \xi. \quad (39)$$

Note that (38) is the separation condition (26) with a margin, evaluated over the sublevel set, whereas (39) is the compatibility condition of [32] for moving sets. In other words, the two conditions couple the initial-error level c and the reference demand to the state envelopes and to the finite actuator authority. A more conservative but easier sufficient test for (39) is $M_c \leq \xi \underline{\mathcal{G}}$, where M_c is the supremum of $\|\mathcal{N}\|_\infty$ over the same set and $\underline{\mathcal{G}}$ is given by (13).

Remark 3. Note that (38) cannot be removed if exact tracking is required, since maintaining $x_1 \equiv y_d$ requires $x_2 = G_1^{-1}(\dot{y}_d - f_1)$, which may leave $\mathbb{X}_2(t)$, and the same holds at every block. Unlike the feasibility condition of BLF backstepping [12], [13], [16], it carries no barrier width, and by Lemma 1 its violation does not violate the state constraints. The design in [18] avoids this condition at the cost of uniform ultimate boundedness. Condition (39) cannot be removed either, since $\mathcal{G}u_c = \mathcal{N}$ identically, so the command cancels the inward vector field on which Theorem 1 relies. Moreover, (39) becomes more restrictive as the actuator set contracts faster, through ε_0, and as the gains k_i increase, through $\dot{\eta}_n$ in $\mathcal{N}$.

Proposition 2 (Local nonvacuity of APC compatibility)**.** *Consider the plant (1) with the APIR (4), the virtual controls (28) and (32), and the command (36), under Assumptions 1 to 4. Suppose that $u_{\max}$ and $u_{\min}$ are bounded, and that there exists $\underline{r} > 0$ such that $r_i(t) \geq \underline{r}\mathbf{1}$, $i = 1, \dots, n$, for all $t \geq 0$. Let $\eta_i^*(t) := \eta_i(\mathbf{0}, t)$, $i = 1, \dots, n$, denote the virtual controls evaluated at $\mathbf{X} = \mathbf{0}$. Moreover, suppose that there exist $\sigma^* \in (\sigma_\eta, 1)$, a compact set $\mathbb{K} \subset \mathbb{R}^m$, and constants $\delta > 0$ and $\xi_0 > 0$ such that the following conditions hold for all $t \geq 0$. First, $\|(\eta_i^*(t) - \check{x}_{i+1}(t))/r_{i+1}(t)\|_\infty \leq 1 - \sigma^*$, $i = 1, \dots, n-1$. Second, $\eta_n^*(t) \in \mathbb{K}$, and every u with $\mathrm{dist}_\infty(u, \mathbb{K}) \leq 2\delta$ lies in $\mathbb{U}(t)$. Third, the zero-error command satisfies*

$$Q_0 := \sup_{t\geq 0} \|\mathcal{G}(\eta_n^*(t), t)^{-1} \mathcal{N}(\mathbf{0}, \eta_n^*(t), t)\|_\infty < \xi_0. \quad (40)$$

Then there exists $c^* > 0$ such that every level $c \in (0, c^*]$ is APC-compatible with $\xi = \xi_0$ and $\mathbb{K}_c := \{u \mid \mathrm{dist}_\infty(u, \mathbb{K}) \leq \delta\}$.

Proof. Let $w(t)$ denote the vector formed by y_d, the state envelopes, the actuator bounds, and the derivatives of these signals that enter (31) and (36). By Assumptions 2 and 3 and the boundedness of $u_{\max}$ and $u_{\min}$, $w(t)$ remains in a bounded set, whose closure we denote by $\mathbb{W}$. By construction, each η_i, the input $u_{\mathbf{X}}$ determined by $\mathbf{X}$, and $\mathcal{N}$ are continuous functions of $(\mathbf{X}, w)$ on the open set $\mathcal{O}$ on which every width in (18) and (19) is positive.

The point $\mathbf{X} = \mathbf{0}$ lies uniformly inside $\mathcal{O}$. Indeed, at $\mathbf{X} = \mathbf{0}$, every z_i vanishes, so that $x_1 = y_d$ and $x_{i+1} = \eta_i^*$. Hence,

the widths of the first block are at least ε_y by Assumption 2. Moreover, since $\alpha_{i+1} = r_{i+1} + (\eta_i^* - \check{x}_{i+1})$ and $\beta_{i+1} = r_{i+1} - (\eta_i^* - \check{x}_{i+1})$ by (19), the first condition gives $\alpha_{i+1} \succeq \sigma^* r_{i+1} \succeq \sigma^* \underline{r}\mathbf{1}$ and, similarly, $\beta_{i+1} \succeq \sigma^* \underline{r}\mathbf{1}$. Consequently, there exists $\rho > 0$ such that $\{\mathbf{X} \mid \|\mathbf{X}\| \le \rho\} \times \mathbb{W} \subset \mathcal{O}$, and the functions above are uniformly continuous on this compact set.

Uniform continuity yields $\rho_1 \in (0, \rho]$ such that, for all $\|\mathbf{X}\| \le \rho_1$ and all $t \ge 0$,

$$\left\| \frac{\eta_i(\mathbf{X}, t) - \eta_i^*(t)}{r_{i+1}(t)} \right\|_\infty \le \sigma^* - \sigma_\eta, \quad i = 1, \dots, n-1, \tag{41}$$

$$\|u_{\mathbf{X}} - \eta_n^*(t)\|_\infty \le \delta, \tag{42}$$

$$\|\mathcal{G}(u_{\mathbf{X}}, t)^{-1}\mathcal{N}(\mathbf{X}, u_{\mathbf{X}}, t) - \mathcal{G}(\eta_n^*, t)^{-1}\mathcal{N}(\mathbf{0}, \eta_n^*, t)\|_\infty \le \xi_0 - Q_0. \tag{43}$$

In (41), the division by $r_{i+1} \succeq \underline{r}\mathbf{1}$ preserves uniform continuity. For (43), note that $u_{\mathbf{X}}$ and η_n^* lie in $\mathbb{K}_c$ by (42) and the second condition, and every point of $\mathbb{K}_c$ lies at a distance of at least δ from the boundary of $\mathbb{U}(t)$. Since $u_{\max}$ and $u_{\min}$ are bounded, the diagonal entries of $\mathcal{G}$ in (5) are bounded below by a positive constant on $\mathbb{K}_c$, so that $\mathcal{G}^{-1}$ is uniformly continuous there.

Finally, since $\mathcal{W} \ge \frac{1}{2}\underline{\mu}\|\mathbf{X}\|^2$ with $\underline{\mu} := \min\{\mu_1, \dots, \mu_n, 1\}$, the choice $c^* := \underline{\mu}\rho_1^2/2$ gives $\|\mathbf{X}\| \le \rho_1$ on Ω_c for every $c \in (0, c^*]$. Hence, by the triangle inequality, the first condition and (41) give $P_c \le (1 - \sigma^*) + (\sigma^* - \sigma_\eta) = 1 - \sigma_\eta$, which is (38). Moreover, (42) gives $u_{\mathbf{X}} \in \mathbb{K}_c$, and (43) together with (40) gives $Q_c(\xi_0) \le Q_0 + (\xi_0 - Q_0) = \xi_0$, which is (39). Therefore, every level $c \in (0, c^*]$ is APC-compatible. □

Note that the hypotheses of Proposition 2 involve only the zero-error closed loop, that is, the reference, the envelopes, and the actuator bounds, so that they can be checked before the controller is run.

We first establish that the state constraints hold without any compatibility condition.

Proposition 3 (State constraints on the interval of existence)**.** *Consider the plant (1) with the APIR (4), the virtual controls (28) and (32), and the command (36), under Assumptions 1 to 4. Let $x_i(0) \in \mathbb{X}_i(0)$, $i = 1, \dots, n$, and $u(0) \in \mathbb{U}(0)$, and let (26) hold at $t = 0$. Then $x_i(t) \in \mathbb{X}_i(t)$, $i = 1, \dots, n$, for all $t \in [0, T_{\max})$, where $[0, T_{\max})$ is the maximal interval of existence of the closed-loop solution.*

Proof. The closed-loop vector field is defined on the open set $\mathcal{D}$ on which every width α_i, β_i is positive, every coordinate z_i is finite, and $\mathcal{G}(u, t)$ is invertible. By Assumption 2, (26) at $t = 0$, and $u(0) \in \mathbb{U}(0)$, the initial condition lies in $\mathcal{D}$. Hence, the solution remains in $\mathcal{D}$ on its maximal interval of existence $[0, T_{\max})$, so that every $z_i(t)$ is finite. By Lemma 1, a finite z_i is equivalent to $\underline{x}_i(t) < x_i(t) < \overline{x}_i(t)$, which yields $x_i(t) \in \mathbb{X}_i(t)$, $i = 1, \dots, n$, for all $t \in [0, T_{\max})$. □

Note that, by Proposition 3, no state leaves its envelope while the solution exists. In other words, if compatibility fails, then either the input leaves $\mathbb{U}(t)$ or the solution ceases to exist in finite time, that is, $T_{\max} < \infty$.

Theorem 4 (Exponential admissibility-preserving tracking)**.** *Consider the plant (1) with the APIR (4), the virtual controls (28) and (32), and the command (36), under Assumptions 1 to 4. Let $x_i(0) \in \mathbb{X}_i(0)$, $i = 1, \dots, n$, and $u(0) \in \mathbb{U}(0)$, and let (26) hold at $t = 0$. Moreover, let the level c be APC-compatible with $\mathcal{W}(0) < c$, and define $\bar{\omega}_1 := \sup_{t \ge 0} \|\overline{x}_1(t) - \underline{x}_1(t)\|_\infty$. Then the closed-loop solution exists for all $t \ge 0$, Ω_c is forward invariant, (37) holds for all $t \ge 0$, and the tracking error satisfies*

$$\|\varphi_1(t)\|_\infty \le \frac{\bar{\omega}_1}{4} \sqrt{\frac{2\mathcal{W}(0)}{\mu_1}} e^{-\vartheta t}. \tag{44}$$

Moreover, for all $t \ge 0$, $x_i(t) \in \mathbb{X}_i(t)$, $i = 1, \dots, n$, the realized input satisfies (12), and hence $u(t) \in \mathbb{U}(t)$, and all closed-loop signals are bounded.

Proof. Let $[0, T)$ denote the maximal interval on which the solution exists with $\mathbf{X}(t) \in \Omega_c$ and $u(t) \in \mathbb{U}(t)$. On this interval, (37) gives $\mathcal{W}(t) \le \mathcal{W}(0) e^{-2\vartheta t} < c$, so that $\mathbf{X}(t)$ remains in the interior of Ω_c. Hence, by (38), every width α_{i+1} and β_{i+1} is at least $\sigma_\eta r_{i+1}$, and every coordinate z_i is defined. Since each z_i is bounded and each width is bounded below, Proposition 3 implies that every state remains at a uniform distance from its envelope. Moreover, compatibility gives $\|u_c(t)\|_\infty \le Q_c(\xi) \le \xi$ on $[0, T)$. Therefore, Theorem 1, applied on this interval, yields (12) and (13), so that u remains at a uniform distance from the boundary of $\mathbb{U}(t)$. Consequently, the closed-loop state is confined to a compact subset of the open domain of the vector field, and the extension theorem [38, Th. 3.3] gives $T = \infty$.

To obtain (44), note from (21) that $q_{1,l} \ge 4/(\alpha_{1,l} + \beta_{1,l})$ on $(-\alpha_{1,l}, \beta_{1,l})$, with equality at the midpoint. Integrating $q_{1,l}$ from 0 to $\varphi_{1,l}$ gives $|z_{1,l}| \ge 4|\varphi_{1,l}|/(\alpha_{1,l} + \beta_{1,l})$. Since $\alpha_{1,l} + \beta_{1,l} = \overline{x}_{1,l} - \underline{x}_{1,l} \le \bar{\omega}_1$ and $\mu_1 \|z_1\|^2 \le 2\mathcal{W} \le 2\mathcal{W}(0) e^{-2\vartheta t}$, (44) follows. Finally, the remaining closed-loop signals are bounded by the smoothness of f_i, G_i, and the virtual controls on the compact set obtained above. □

Corollary 1 (Explicit state margins and input rate)**.** *Consider the plant (1) with the APIR (4), the virtual controls (28) and (32), and the command (36), under Assumptions 1 to 4. Suppose that the initial conditions satisfy $x_i(0) \in \mathbb{X}_i(0)$, $i = 1, \dots, n$, $u(0) \in \mathbb{U}(0)$, and (26) at $t = 0$, and that the level c is APC-compatible with $\mathcal{W}(0) < c$. For $i = 1, \dots, n$, define $\bar{z}_i(t) := \sqrt{2\mathcal{W}(0)/\mu_i}\, e^{-\vartheta t}$, and let the relative margin of the lth component of x_i be $m_{i,l} := \min\{x_{i,l} - \underline{x}_{i,l}, \overline{x}_{i,l} - x_{i,l}\}/(\overline{x}_{i,l} - \underline{x}_{i,l})$. Moreover, let $\bar{\omega}_1 := \sup_{t \ge 0} \|\overline{x}_1(t) - \underline{x}_1(t)\|_\infty$. Then, for every component l and all $t \ge 0$,*

$$m_{1,l}(t) \ge \frac{\varepsilon_y}{\varepsilon_y + (\bar{\omega}_1 - \varepsilon_y) e^{\bar{z}_1(t)}}, \tag{45}$$

$$m_{i,l}(t) \ge \frac{\sigma_\eta}{\sigma_\eta + (2 - \sigma_\eta) e^{\bar{z}_i(t)}}, \quad i \ge 2. \tag{46}$$

where ε_y and σ_η are given in Assumption 2 and (38), respectively. If, in addition, $u_{\max}$ and $u_{\min}$ are bounded, then the realized input satisfies

$$\|\dot{u}(t)\|_\infty \le p_1 (\xi + p_2 \bar{u}), \tag{47}$$

where ξ is the command bound in (39), $\bar{u} := \max\{\bar{s} \sup_{t,l} u_{\max,l}(t), \bar{\zeta} \sup_{t,l} |u_{\min,l}(t)|\}$, and $\bar{s}$ and $\bar{\zeta}$ are given in (11).

Proof. From the definition of $\mathcal{W}$ and Theorem 4, for every $i = 1, \dots, n$, $\frac{1}{2}\mu_i \|z_i(t)\|^2 \le \mathcal{W}(t) \le \mathcal{W}(0)e^{-2\vartheta t}$,, where the second inequality follows from (37) by the comparison lemma. Hence, $|z_{i,l}(t)| \le \|z_i(t)\| \le \bar{z}_i(t)$ for every component l.

Next, from (17), (18), and (19), one has $x_{i,l} - \underline{x}_{i,l} = \alpha_{i,l} + \varphi_{i,l}$ and $\bar{x}_{i,l} - x_{i,l} = \beta_{i,l} - \varphi_{i,l}$. Substituting the inverse (25) gives

$$x_{i,l} - \underline{x}_{i,l} = \frac{\alpha_{i,l}(\alpha_{i,l} + \beta_{i,l})e^{z_{i,l}}}{\beta_{i,l} + \alpha_{i,l}e^{z_{i,l}}}, \tag{48}$$

$$\bar{x}_{i,l} - x_{i,l} = \frac{\beta_{i,l}(\alpha_{i,l} + \beta_{i,l})}{\beta_{i,l} + \alpha_{i,l}e^{z_{i,l}}}, \tag{49}$$

where $\alpha_{i,l}+\beta_{i,l} = \bar{x}_{i,l} - \underline{x}_{i,l}$. Dividing (48) and (49) by $\alpha_{i,l}+\beta_{i,l}$, and suppressing the indices, the relative margin becomes $m = \min\left\{\frac{\alpha}{\alpha+\beta e^{-z}}, \frac{\beta}{\beta+\alpha e^{z}}\right\}$. The first term increases and the second decreases with z. Since $-\bar{z}_i \le z \le \bar{z}_i$, evaluating the first term at $z = -\bar{z}_i$ and the second at $z = \bar{z}_i$ yields

$$m \ge \min\left\{\frac{\alpha}{\alpha + \beta e^{\bar{z}_i}}, \frac{\beta}{\beta + \alpha e^{\bar{z}_i}}\right\}. \tag{50}$$

Writing $\beta = \omega - \alpha$ with $\omega := \alpha + \beta$, the first term in (50) equals $\alpha/(\alpha + (\omega - \alpha)e^{\bar{z}_i})$, which increases with α and decreases with ω, since $e^{\bar{z}_i} \ge 1$. The same holds for the second term with α and β interchanged. Therefore, if $\alpha \ge \underline{w}$, $\beta \ge \underline{w}$, and $\omega \le \bar{\omega}$, then

$$m \ge \frac{\underline{w}}{\underline{w} + (\bar{\omega} - \underline{w})e^{\bar{z}_i}}. \tag{51}$$

For $i = 1$, Assumption 2 and (18) give $\alpha_{1,l} \ge \varepsilon_y$ and $\beta_{1,l} \ge \varepsilon_y$, and $\alpha_{1,l} + \beta_{1,l} = \bar{x}_{1,l} - \underline{x}_{1,l} \le \bar{\omega}_1$. Hence, (51) with $\underline{w} = \varepsilon_y$ and $\bar{\omega} = \bar{\omega}_1$ gives (45). For $i \ge 2$, (19) gives $\alpha_{i,l} = r_{i,l} + (\eta_{i-1,l} - \check{x}_{i,l})$ and $\beta_{i,l} = r_{i,l} - (\eta_{i-1,l} - \check{x}_{i,l})$. Since Ω_c is forward invariant by Theorem 4, (38) gives $|\eta_{i-1,l} - \check{x}_{i,l}| \le (1 - \sigma_\eta) r_{i,l}$, so that $\alpha_{i,l} \ge \sigma_\eta r_{i,l}$, $\beta_{i,l} \ge \sigma_\eta r_{i,l}$, and $\alpha_{i,l} + \beta_{i,l} = 2r_{i,l}$. Hence, (51) with $\underline{w} = \sigma_\eta r_{i,l}$ and $\bar{\omega} = 2r_{i,l}$ gives (46).

Finally, by (4), $\dot{u}_l = p_1 \mathcal{S}_l u_{c,l} - p_1 p_2 u_l$, where $0 < \mathcal{S}_l \le 1$ along the solution by (12). Hence, $|\dot{u}_l| \le p_1 |u_{c,l}| + p_1 p_2 |u_l|$. Since Ω_c is forward invariant, (39) gives $\|u_c\|_\infty \le \xi$, and (12) gives $|u_l| \le \bar{u}$, which yields (47). □

Note that, as $t \to \infty$, the bounds in (45) and (46) approach $\varepsilon_y/\bar{\omega}_1$ and $\sigma_\eta/2$, respectively, since $\bar{z}_i(t) \to 0$.

Remark 4. Note that Theorem 4 extends the result of [32] to full state constraints and moving sets. Since (37) is exact, the tracking error converges to zero for a time-varying reference without any boundary-layer residual. However, the design requires the analytic expansion of $\dot{\eta}_n$, which grows with n, and smoothness of the reference and of every envelope. Relaxing (38) by filtering the recursion, as in [18], would yield uniform ultimate boundedness instead of exponential convergence.

Corollary 2 (Transfer of input safety under realization error)**.** *Let the input applied by the physical actuator be $u_a = u + d_u$, where the realization error satisfies $\|d_u(t)\|_\infty \le \bar{d}_u$ for all $t \ge 0$. Let the APIR (4) be designed on the tightened interval $\widetilde{\mathbb{U}}(t) := (u_{\min}(t) + \bar{d}_u \mathbf{1}, u_{\max}(t) - \bar{d}_u \mathbf{1})$, with a piecewise-continuous command satisfying $\|u_c(t)\|_\infty \le \xi$ for all $t \ge 0$. Moreover, let the tightened bounds satisfy Assumption 4, that is, $u_{\max,l} - \bar{d}_u \ge \underline{u}_{\max} > 0$, $u_{\min,l} + \bar{d}_u \le -\underline{u}_{\min} < 0$, and*

$$\frac{\dot{u}_{\max,l}}{u_{\max,l} - \bar{d}_u} \ge -p_1 p_2 + \varepsilon_0,$$

$$\frac{\dot{u}_{\min,l}}{u_{\min,l} + \bar{d}_u} \ge -p_1 p_2 + \varepsilon_0, \tag{52}$$

for every component l and all $t \ge 0$. Then $u_{\text{phys}}(t) \in \mathbb{U}(t)$ for all $t \ge 0$.

Proof. Applying Theorem 1 to the tightened bounds yields $u(t) \in \widetilde{\mathbb{U}}(t)$ for all $t \ge 0$. Since $\|d_u(t)\|_\infty \le \bar{d}_u$, it follows that $u_a(t) = u(t) + d_u(t) \in \mathbb{U}(t)$. □

Note that (52) does not follow from (8). Since tightening divides the rate of a contracting boundary by a smaller positive number, bounds that satisfy the rate condition on $\mathbb{U}(t)$ may violate it on $\widetilde{\mathbb{U}}(t)$. Hence, the margin ε_0 must be large enough to absorb the factor $u_{\max,l}/(u_{\max,l} - \bar{d}_u)$, so that tightening reduces both the available authority and the rate margin.

IV. Numerical Simulations

In this section, we demonstrate the performance of the proposed time-varying APC strategy through numerical simulations. We first consider a scenario in which the state envelopes translate over time while the actuator set oscillates and varies in width. We then investigate the effect of the control gains on the performance of the proposed strategy. Thereafter, we consider a scenario in which the actuator set contracts at different rates, including rates reaching 95% of the admissible rate $p_1 p_2$ specified in Assumption 4. Finally, we investigate the sensitivity of the design to the choice of weights μ_i under both oscillating and contracting actuator sets. In all cases, we consider the plant (1) with $n = 2$ and $m = 2$, where

$$f_1 = \begin{bmatrix} 0.5 \sin x_{1,1} \\ -0.3x_{1,2} + 0.2x_{1,1}^2 \end{bmatrix}, \; G_1 = \begin{bmatrix} 1.2 + 0.1x_{1,1}^2 & 0.2 \\ -0.1 & 1.0 \end{bmatrix},$$

$$f_2 = \begin{bmatrix} 0.4x_{2,1}x_{1,1} \\ -0.5x_{2,2} + 0.3 \sin x_{1,2} \end{bmatrix}, \; G_2 = \begin{bmatrix} 1.0 & 0.3 \\ 0.2 & 1.1 \end{bmatrix}.$$

The reference trajectory is chosen as $y_d = [0.5 \sin 0.5t, \; 0.4 \cos 0.4t]^\top$. The state envelopes are defined by $\underline{x}_1 = (-1 + 0.3 \sin 0.6t)\mathbf{1}$, $\bar{x}_1 = (1 + 0.3 \sin 0.6t)\mathbf{1}$, $\underline{x}_2 = (-2.5 + 0.4 \sin 0.35t)\mathbf{1}$, and $\bar{x}_2 = (2.5 + 0.4 \sin 0.35t)\mathbf{1}$. The actuator bounds are given by $u_{\max} = (3 + \sin 0.3t)\mathbf{1}$ and $u_{\min} = -(2.5 + 0.8 \cos 0.25t)\mathbf{1}$. Thus, both state envelopes translate over time while maintaining constant widths, whereas the actuator set has simultaneous translation and width variation.

The APIR parameters are selected as $p_1 = 5$, $p_2 = 0.5$, and $\gamma = 2$, which yield $p_1 p_2 = 2.5$. Since $\sup_t |\dot{u}_{\max,l}/u_{\max,l}| \le 0.15$ and $\sup_t |\dot{u}_{\min,l}/u_{\min,l}| \le 0.118$, Assumption 4 holds. The controller gains are chosen as $k_1 = k_2 = 2$ and $k_3 = 4$, such that $\vartheta = 2$, and $\sigma_\eta = 0.05$. The weights are selected as $\mu_i = \bar{r}_i^2/4$, where $\bar{r}_1 = 1$ and $\bar{r}_2 = 2.5$ denote the half-widths of the respective state envelopes. This implies $\mu_1 = 0.25$ and $\mu_2 = 1.5625$. The initial conditions are chosen as $x_1(0) = [0.35, -0.25]^\top$, $x_2(0) = [-0.501, 1.238]^\top$, and $u(0) = [0.436, -0.680]^\top$. To quantify constraint satisfaction, we define the relative margin of a component v_l with respect to its admissible interval as

$$\mathcal{M}_{v,l} := \inf_{t \ge 0} \frac{\min\{v_l(t) - \underline{v}_l(t), \; \bar{v}_l(t) - v_l(t)\}}{\bar{v}_l(t) - \underline{v}_l(t)},$$

and, for a vector signal, take the minimum margin among its components. Note that a positive margin indicates that the signal remains strictly inside its admissible set throughout the interval.

The performance of the proposed strategy is presented in Figure 3. It can be observed from Figures 3a and 3e that both output states converge to their respective reference trajectory within a brief transient of approximately 1.5 s. During this transient, the controller generates a larger commanded input to nullify the tracking errors. As shown in Figure 3d, these commands decrease to near zero once the tracking errors have converged to zero. Correspondingly, the realized inputs in Figure 3c settle near zero and remain strictly within the actuator set throughout the time, despite the time-varying actuator bounds and the changes in the width. Furthermore, Figures 3a and 3b show that both state blocks remain strictly inside their respective moving envelopes, with relative margins of 0.302 and 0.252, respectively. The relative margin of the realized input is 0.327. One may observe from Figure 3f that the normalized virtual control remains well within the separation bound $\pm(1-\sigma_\eta) = \pm 0.95$. This, in turn, verifies that the APC compatibility condition is satisfied throughout the simulation.

We now investigate the effect of the control gains on the performance of the proposed APC framework. The gains are chosen from the set $k_1 = k_2 = k \in \{1.5, 2, 2.5, 3\}$ with $k_3 = 4$, while the other simulation settings are kept the same to those used in Figure 3. The performance of the proposed strategy is summarized in Table I. It can be observed that the state and input constraints are satisfied in every case. Furthermore, as the gain increases from 1.5 to 3, the peak of the normalized virtual control increases almost linearly from 0.315 to 0.675, while remaining below the separation bound $1-\sigma_\eta = 0.95$. Thus, the separation condition is satisfied throughout the gain variations. On the other hand, the input margin decreases from 0.383 to 0.014, while the peak commanded input increases from 0.8 to 31.3. This is primarily due to the fact that the numerator of the commanded input contains the term $\dot{\eta}_n$, which increases with the controller gains, as discussed in Remark 3.

TABLE I: Effect of the control gains.

k	$\max\|(\eta_1 - \check{x}_2)/r_2\|_\infty$	$\mathcal{M}_{x_1}$	$\mathcal{M}_{x_2}$	$\mathcal{M}_u$	$\max\|u_c\|_\infty$
1.5	0.315	0.302	0.312	0.383	0.8
2.0	0.435	0.302	0.252	0.327	1.7
2.5	0.555	0.302	0.192	0.189	3.5
3.0	0.675	0.302	0.132	0.014	31.3

We next consider a scenario where the actuator set is contracting, which may arise from actuator derating or a fault that reduces the available actuator authority. The actuator bounds are chosen as $u_{\max}(t) = [1 + 2e^{-at}]\mathbf{1}$ and $u_{\min}(t) = -[0.8 + 1.7e^{-at}]\mathbf{1}$, such that the admissible interval contracts from $(-2.5, 3)$ to $(-0.8, 1)$ at a rate governed by a. The magnitude of the lower bound contracts faster in logarithmic terms, with $\sup_t |\mathrm{d}\ln|u_{\min,l}|/\mathrm{d}t| = 0.68a$, attained at $t = 0$. We consider $a \in \{1, 2, 3, 3.5\}$, for which the maximum contraction rate remains below $p_1 p_2 = 2.5$. Thus, Assumption 4 is satisfied, with $\varepsilon_0 = 0.12$ for $a = 3.5$. The initial input is chosen as $u(0) = [0.406, -0.710]^\top$, while all other simulation settings are kept the same as in Figure 3.

TABLE II: Effect of the contraction rate.

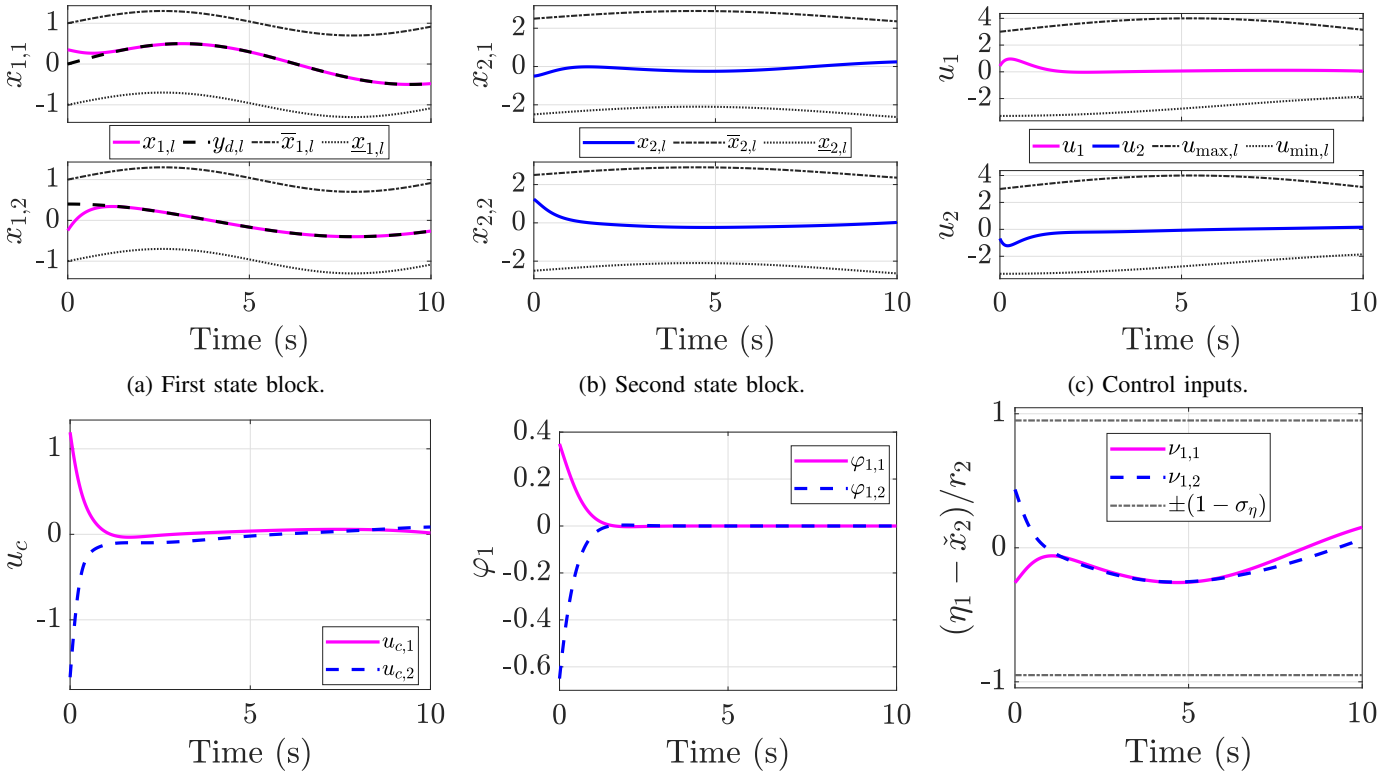
a	$\sup_t \|\mathrm{d}\ln\|u_{\min}\|/\mathrm{d}t\|$	$\mathcal{M}_u$	$\max\|u_c\|_\infty$
1.0	0.680	0.194	1.72
2.0	1.360	0.142	1.72
3.0	2.040	0.084	1.72
3.5	2.380	0.056	1.72

The performance of the proposed strategy in terms of the effect of the contraction rate is summarized in Table II, while the response for fastest contraction, $a = 3.5$, is presented in Figure 4. It can be observed from Table II that the input margin decreases monotonically from 0.194 to 0.056 as the contraction rate increases from 0.68 to 2.38. However, it remains positive for all values of a, as guaranteed by Theorem 1. Moreover, the peak commanded input remains the same at 1.72, which is attained at $t = 0$, and the tracking error converges to zero, since the interval $(-0.8, 1)$ still contains the input required to track the desired reference trajectory. Furthermore, Figures 4a, 4b, 4e and 4f show that even for the fastest contraction, both state blocks remain within their respective envelopes, and the tracking errors converge to zero by exhibiting a similar behavior to those in Figure 3. Thus, the proposed framework accommodates the actuator contraction without increasing the peak command or degrading the tracking accuracy.

Finally, we assess the sensitivity of the proposed design to the choice of weights μ_i. We consider the same two scenarios: the actuator set oscillates and varies in width, and the actuator set contracts. The state and input envelopes are chosen the same as in Figure 3 and in Figure 4 (with $a = 3$ and $a = 3.5$). The value of Lyapunov weights are chosen from the set $\mu_1 \in 0.0625, 0.125, 0.25, 0.5$, spanning a factor of eight around $\bar{r}1^2/4 = 0.25$, while keeping $\mu_2 = 1.5625$ fixed. By keeping the other settings the same as those in Figure 3, we summarize the performance in Table III. In the table, nominal means the case in which the actuator set oscillates, while $a = 3$ and $a = 3.5$ correspond to the contracting actuator set. It can be observed that the state and input constraints are satisfied in each case. As μ_1 decreases from 0.25 to 0.0625, the peak value of η_2 increases from 1.20 to 1.36, while the input margins change by less than seven percent and the peak command by less than two percent. In contrast, increasing μ_1 to 0.5 leaves the response for the case when the actuator set oscillates and varies in width practically unchanged but reduces the input margins to 0.054 and 0.024 for $a = 3$ and $a = 3.5$, respectively. This is because the cross term in (32) increases with μ_1/μ_2, making its effect on the commanded input more pronounced as the actuator set contracts. Thus, the choice $\mu_i = \bar{r}_i^2/4$ lies near the upper end of a broad range of acceptable weights.

V. Conclusions

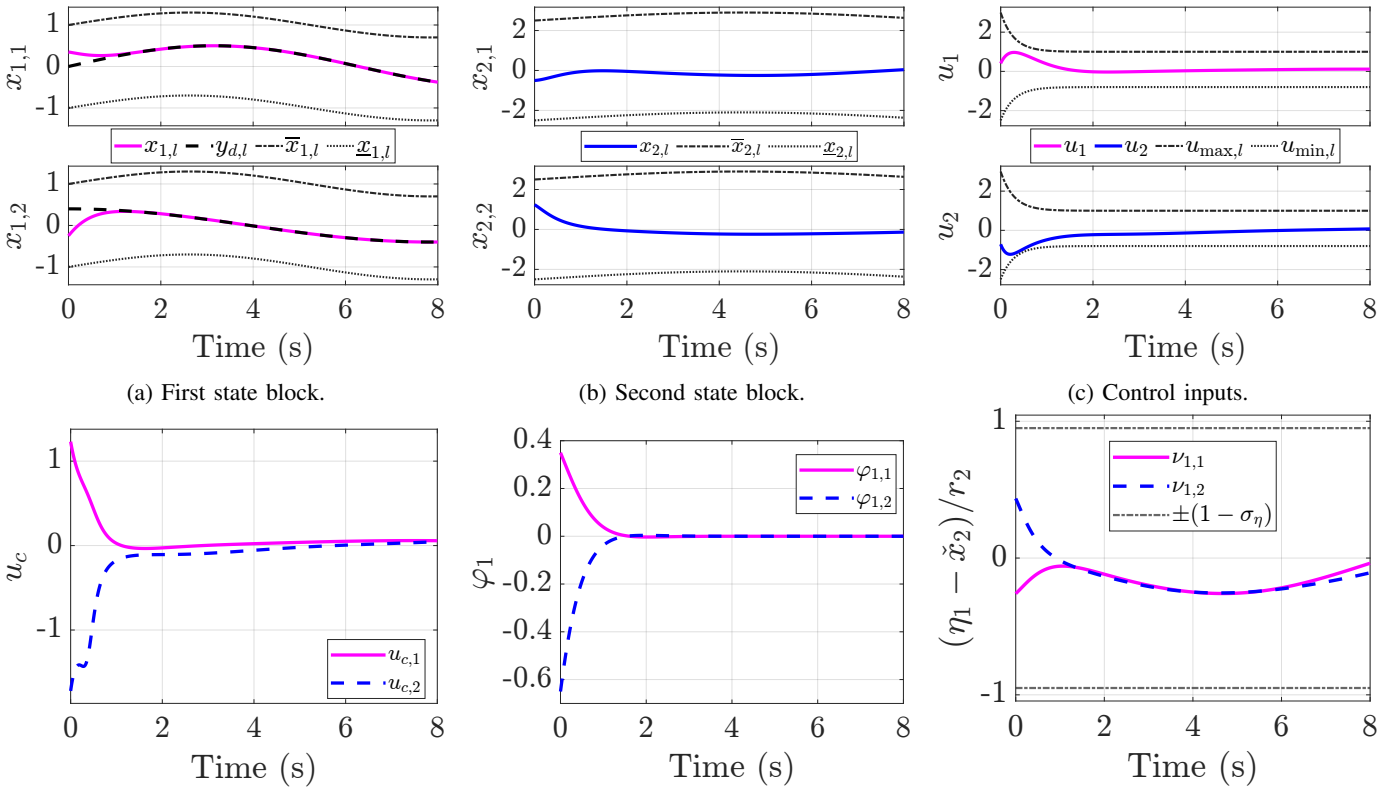
This work developed a time-varying APC framework for block strict-feedback nonlinear systems subject to time-varying asymmetric actuator and full state constraints. The



(a) First state block. (b) Second state block. (c) Control inputs.

(d) Commanded inputs. (e) Tracking errors. (f) Normalized virtual controls.

Fig. 3: Full-state constrained tracking under an oscillating actuator set.



(a) First state block. (b) Second state block. (c) Control inputs.

(d) Commanded inputs. (e) Tracking errors. (f) Normalized virtual controls.

Fig. 4: Full-state constrained tracking under a contracting actuator set.

TABLE III: Effect of the Lyapunov weights.

	$\mathcal{M}_u$				
μ_1	nominal	$a = 3$	$a = 3.5$	$\max\|\eta_2\|_\infty$	$\max\|u_c\|_\infty$
0.0625	0.306	0.084	0.058	1.36	1.70
0.125	0.314	0.086	0.059	1.30	1.68
0.25	0.327	0.084	0.056	1.20	1.67
0.5	0.322	0.054	0.024	1.10	1.74

proposed APIR keeps the realized input inside a moving actuator set whenever each bound contracts more slowly than the decay rate of the realization, with a closed-form lower bound on its gain. By applying the logarithmic barrier coordinate at every block of the backstepping recursion, the full state constraints hold without barrier widths reserved for the errors, and the feasibility condition of BLF backstepping is replaced by a separation condition on the virtual controls. Under APC compatibility, the tracking error converges exponentially to zero, while every state and the realized input remain inside their moving sets for all time.

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